Basic SEO and Career-Keyword Analysis

Portfolio: akthernaimudheen.in
Research date: 22 October 2026
Scope: Public-source and on-site analysis only. No website, repository, or public profile was changed.

Executive summary

The portfolio already has a strong, specific foundation for job discovery: it identifies an AI application/full-stack engineering focus, links to project case studies, and gives unusually concrete explanations of architecture, testing, and limitations. The highest-priority issue found is technical: the original HTML response for multiple routes emits the homepage canonical URL, while the browser-rendered versions change those canonicals to each route. Google advises that JavaScript should not change a canonical URL specified in the original HTML; correct this at the server-rendered/source level and verify it in Search Console. The project and article pages also have useful individual titles and content after rendering, so this is a conflicting-signal issue rather than a conclusion that Google cannot see the pages. A local simulated mobile Lighthouse run scored 73/100 for performance and 92/100 for SEO, with a 4.0-second lab LCP; these are single-run lab results, not field Core Web Vitals. Free public research did not establish trustworthy India-specific search volumes, keyword difficulty, rankings, or traffic. Job listings indicate role vocabulary and hiring activity, not Google query demand. Prioritize canonical consistency, distinct indexable page metadata, then validate query demand and indexation with Search Console.

Method and limits

I inspected the public homepage, About, Work, two project pages, a technical article, the blog route, robots.txt, and the sitemap. I also checked the raw HTTP HTML and the JavaScript-rendered DOM with a local Chromium browser, and ran Lighthouse 12.8.2 locally using its simulated mobile profile on the homepage. Public web search, official Google Search documentation, public portfolios, and employer career pages supplemented the site review.

Semrush is connected, but its reports could not run because the account needs additional API units. I did not retry. Google PageSpeed Insights' public API also returned a quota-exceeded response in this session, so the performance snapshot below is from local Lighthouse instead. No Search Console or analytics data was available. I therefore report **no search-volume, KD, traffic, ranking, or conversion numbers**. Every keyword in the table has unmeasured demand and difficulty. Public snippets, portfolio guides, and job postings

are qualitative evidence only. Job-board listings can expire or change and must not be treated as Google search-volume estimates.

What the public site currently communicates

The homepage positions Akther Naimudheen as an AI Engineer and Full-Stack Developer in Kerala, India, and emphasizes generative AI pipelines, clinical documentation quality assurance, and web systems. The Work index separates AI applications, full-stack systems, and client delivery. The iScribe case study explains its synthetic-data-only scope, validation pipeline, human review boundary, and technical choices. PromptToShort highlights Python/FastAPI, a media pipeline, and a stated 160+ automated tests. The published clinical-AI article adds a detailed technical narrative and explicitly states that the work was evaluated on synthetic/sanitized material, not live clinical deployment. These are strong *on-page proof points as presented on the site*; this public review did not independently verify the underlying code, benchmarks, employment history, or project outcomes. [1] [2] [3] [4] [5]

The clearest differentiation is not the generic label "AI engineer." It is the combination of applied AI systems, verification/testing, and transparent limits. Keep that proof-led angle, while making it easy for a recruiter to understand the actual target role and seniority.

Technical and page-level findings

Priority	Finding from this check	Why it matters	Recommended next step
P1	In the original HTTP HTML, the homepage, About, Work, project, and article routes all return the homepage canonical. In the browser-rendered DOM, About, iScribe, and the article have route-specific canonicals; JavaScript is changing the original value.	This is a conflicting canonical signal. Google says the canonical set in JavaScript should not differ from the canonical in the original HTML. The sitemap lists these pages individually, which is another signal to reconcile. This is not proof that Google currently selected the wrong canonical or excluded a page.	Emit the correct self-referencing canonical for each indexable route in the initial HTML/server response. Keep it identical after rendering, and align internal links and sitemap URLs. Validate several URLs in Search Console URL Inspection after deployment. [6] [8] [9]

P1	The sitemap currently lists 11 URLs, including the article and project routes, and uses the `www` hostname. Routes tested returned successful responses. The non-`www` homepage request redirects to the `www` site.	The hostname choice and sitemap are broadly consistent, but per-route canonicals in the raw HTML conflict with the sitemap's page-level URLs. A sitemap is a weaker canonical hint than `rel="canonical"`.	Preserve one HTTPS + `www` URL convention and ensure every sitemap URL returns 200 and a matching canonical. Re-submit/inspect the sitemap in Search Console. 6 9
P2	The raw response repeats the homepage title, description, and H1 on other routes. In the rendered browser DOM, the inspected About, iScribe, and article pages have specific titles, H1s, and canonicals.	Google can render JavaScript and supports JavaScript-set titles/descriptions, but the raw source is less robust for other crawlers and previews. The canonical is the more serious inconsistency because Google specifically advises against changing it from the initial HTML value.	Prefer server-rendered/prerendered route metadata and content. If using client rendering, at minimum make the initial source and rendered metadata consistent. Give every indexable page a unique descriptive title, description, H1, and canonical. 8 10
P2	`/blog` rendered the same article as `/blog/building-clinical-ai-beyond-transcription`; both rendered with the article title/H1 and the article URL as canonical.	This appears to be a blog entry point that resolves directly to the current article, not a separate listing page. With one public article this may be intentional; as the library grows, a true index would improve navigation and internal linking.	Either keep `/blog` as a deliberate redirect/alias to the article, or make it a distinct blog index with its own title, H1, canonical, and links to posts. Do not leave two URL paths with ambiguous behavior.
P3	Local simulated-mobile Lighthouse (homepage, one run on 22 Oct 2026) reported performance **73/100**, SEO **92/100**, LCP **4.0 s**, FCP **2.6 s**,	This is a lab diagnostic under simulated network/device conditions. It is not field-user performance, CrUX data, or a stable	Inspect the identified blocking resources and JS bundle; measure repeat runs and representative project/article routes. Use real-user/Search

	TBT **410 ms**, and CLS **0**. Lighthouse identified render-blocking resources with an estimated 670 ms saving and 76 KiB of unused JavaScript.	guarantee. The opportunity estimates are Lighthouse estimates, not expected gains.	Console or CrUX data where available before claiming Core Web Vitals status.
P3 / monitor	Lighthouse flagged robots.txt as invalid. The fetched file allows general crawling and includes a sitemap directive; its opening contains Cloudflare `Content-Signal` directives. Cloudflare documents that these newer signals may trigger a Search Console “Syntax not understood” warning and says it has observed no crawl-rate/SEO impact from such reports.	Treat this as a parser warning to monitor, not evidence that the site is blocked from Google. Lighthouse separately passed its “page isn’t blocked from indexing” audit.	Confirm the Google Search Console robots.txt report, if available. Do not remove Cloudflare’s content signals solely because Lighthouse reports them; first check the report and desired bot policies. 7 19 20
Positive	The rendered homepage passed Lighthouse checks for title, description, crawlability, descriptive links, image alt attributes, and canonical validity; rendered page routes have useful distinct content.	The site has sound visible content and navigation foundations. Passing a Lighthouse check does not prove indexation, rankings, or accessibility in all contexts.	Preserve the project/case-study depth while fixing source/render consistency and validating indexation.

The local Lighthouse SEO score was reduced by the robots.txt parser check (0/1); the other scored SEO checks in that run passed. Structured data was not scored by Lighthouse. This review did not validate rich-result eligibility or every page’s schema. No mobile field data was available. 7 20

Keyword strategy and page mapping

Demand and difficulty: no credible public source consulted here supplied India-specific search volume and period or a reliable KD figure for these phrases. KD is therefore **unavailable** for all candidates, not "low," "medium," or "high." The intent labels below are inferred from wording and observed public page types. Competition notes are qualitative SERP/page-type observations, not a tool-derived difficulty score.

Candidate phrase	Likely intent and audience	Best destination	Priority / fit
Professional identity			
software developer portfolio India	Informational examples / recruiter review; developers and hiring teams	Homepage + Work index	High fit; broad portfolio phrasing
AI engineer portfolio	Examples / candidate evaluation	Homepage + AI project case studies	High fit if AI work is substantiated
AI application engineer	Role discovery / candidate identity	Homepage headline, About, relevant projects	High fit if this matches target role and experience
software engineer portfolio	Portfolio examples / recruiter review	Homepage + project index	High fit; broad but relevant
Python developer portfolio	Project examples / hiring review	Python project case studies, then Work index	Medium-high; only if specific Python work is shown
Career and role intent			
AI software engineer India	Job/profile discovery	Homepage + About + relevant case studies	High relevance; exact demand unmeasured
AI/ML software engineer India	Employment discovery	Homepage + AI projects	High if both AI/ML and software work are actually evidenced
Python developer India	Job search / candidate discovery	About/skills + Python case studies	Medium-high; broad role competition qualitatively likely
junior AI engineer India	Entry-level job search	Homepage only if seniority is accurate; otherwise omit	Conditional; avoid mislabeling level
entry-level software engineer India	Early-career job search	Homepage/About if accurate	Conditional; broad and seniority-specific

AI application developer	Applied-AI role discovery	AI application project pages	High if case studies demonstrate application engineering
software engineer Kochi	Local job discovery	Homepage/About location line, if accurate	High local relevance if available for Kochi roles
AI engineer Kerala	State-level role discovery	Homepage/About and AI work, if accurate	High local relevance; do not create a thin location page
Projects and technical credibility			
AI application development projects	Project ideas/examples	Work index + individual AI case studies	High as a destination theme
Python software engineering projects	Project examples / learning	Python project case studies	High where repositories/demos and contribution detail exist
building AI applications with Python	How-to / implementation	Technical article or project write-up	Medium-high; tutorial intent is distinct from job intent
AI engineering portfolio projects	Career examples / candidate proof	Work index + project detail pages	High for project discovery, but candidate-seeking intent also exists
software engineering case study	Informational / technical evaluation	Individual case studies	High as a format, not necessarily a volume target
AI application testing and evaluation	Technical methodology / role research	Case study or engineering article with actual evaluation evidence	High differentiation if test artifacts are real and reproducible
debugging and testing software applications	How-to / skill proof	Engineering practices section within project case studies	Medium-high; avoid generic listicle content
Technical writing and security			

AI engineering experiments	Research/learning intent	Experiments index and individual write-ups	Medium; publish only original, reproducible work
software testing and reliability	How-to / employer skill evaluation	Testing/reliability case study	Medium-high when supported by concrete artifacts
responsible vulnerability disclosure	Policy/process information	Security policy page only if an actual channel/process exists	Low/conditional for the current career focus
security assessment methodology	Technical/security research	Authorized assessment methodology/case study	Conditional; avoid implied security credentials without evidence
technical project case study	Informational / recruiter evaluation	Individual project detail pages	High as a portfolio content pattern
Comparative / competitive phrases			
best AI engineer in India	Commercial comparison / hiring shortlist	Do not target as a self-claim; use proof-led homepage instead	Avoid “best” claim; weak direct fit for job search
top AI engineers in India	Rankings / talent discovery	Do not claim rank; ensure name, role, and projects are clear	Avoid unsupported “top” language
AI engineer portfolio India	Portfolio example / recruiter discovery	Homepage and AI Work section	High fit, but exact demand is unmeasured
software developer personal website India	Broad portfolio/design discovery	Homepage only, naturally	Medium; less precise than AI/software role themes

Observed qualitative evidence: portfolio directories and guides show that example/how-to pages are a visible result type for portfolio phrases, but they do not provide a trustworthy India phrase-level demand measure. Public India job descriptions mention Python, React, AI-enabled solutions, testing, deployment, evaluation, and reliability; those terms help validate role relevance, not search volume. For example, Cargill’s Bengaluru Software Engineer (Python + React) listing describes AI-enabled software, testing, reliability, Python/React, and a minimum four years of related experience; Microsoft’s India jobs page showed a Principal Software Engineer and a Senior Researcher role at the

time checked; Sarvam’ s careers page displayed engineering and evaluation roles across Bengaluru/Delhi. These are examples, not a representative labor-market survey, and the seniority requirements differ. [12] [13] [14]

Practical priority order:

1. Lead with a truthful role/location identity and link the homepage to résumé, contact, Work, and GitHub/LinkedIn.
2. Use the Work index to direct recruiters to the strongest two or three relevant projects, rather than trying to rank the homepage for every phrase.
3. Make each case study answer: problem, personal contribution, architecture, tests/evaluation, real limitations, and demo/code access where available.
4. Put project-specific technical queries on their own substantive case-study/article pages. The existing clinical-AI article is already a useful example; build adjacent pieces only when they add new information rather than splitting the same topic into thin pages.
5. Deprioritize “best/top” phrases, generic AI listicles, and location/seniority pages without real unique content.

Competitor and reference-page comparison

These are illustrative public portfolios selected for relevant positioning and structure, not a claim that they rank for the target phrases or compete for the same audience.

Reference	What it presents publicly	Useful lesson for this portfolio
Nithin Pradeep	AI/software identity, selected projects, career timeline, tools, certifications, testimonials, and outbound profiles. Claims are self-published and were not independently verified here.	Recruiters can scan work, role history, skills, and external proof from one page. Use the structural pattern, not unsupported claim volume or identical sections.
Akshay	A differentiated health-tech/creative-tech position with products, experience, projects, GitHub, and research/publication sections. The site presents quantified claims and credentials; this review did not verify them.	Strong portfolios connect a technical specialty to shipped work and provide deeper evidence links. Do not imitate seniority, metrics, patents, or outcomes that are not yours.
Current portfolio	AI pipeline, full-stack systems, and client-delivery categories with separate case studies; the	Keep the clear system/evaluation angle. Surface measurable proof and

	iScribe page includes explicit synthetic-data and clinician-review disclosures.	scope disclosures near the top, link artifacts where public, and avoid presenting a self-directed prototype as clinical deployment.

Competitor traffic, backlinks, rankings, revenue, and traffic share are **not available** from this review and have not been estimated. The observed competing page types include portfolio directories, individual portfolios, employer career pages, and marketplaces; no evidence supports a numerical competitor gap or a "top 20" traffic comparison. [15] [16] [17] [18]

Homepage positioning suggestions

These are copy options for future consideration only; nothing has been changed.

- **Title:** `Akther Naimudheen | AI Application Engineer in Kerala, India`
- **Meta description:** `Explore Akther Naimudheen's AI application and full-stack engineering projects, including clinical AI validation, Python automation, software testing, and technical case studies.`
- **H1:** `AI application engineer building reliable software systems`
- **Supporting line:** `Full-stack applications, AI pipelines, and engineering quality—documented through project architecture, validation, and clear limitations.`

The current rendered homepage already communicates AI engineering, full-stack development, location, and pipeline validation. These options tighten role/location specificity; implement only if they accurately match current goals. Keep each route's title and description unique and descriptive, avoid repeated keyword variants, and make the visible H1 agree with the page topic. Google may construct title links using the title, visible main title, headings, and other sources. [1] [10]

Ten article ideas, ranked by fit and feasibility

Ranking is editorial priority, not a volume or traffic forecast.

1. **How I tested an AI pipeline against unsupported facts, missing evidence, and schema violations** — extend the existing clinical-AI article only with new, reproducible test cases and clear limits; avoid duplicating the existing article.
2. **What 160+ automated tests cover in an AI video-generation pipeline** — explain test categories, failure handling, timing assertions, and which claims can be checked in a public repo/demo.

3. **A practical evaluation harness for AI application features** — define input fixtures, expected outputs, pass/fail rules, regressions, and failure analysis; publish only artifacts that exist.
4. **Designing schema validation and human review boundaries for structured AI output** — focus on implementation choices and known limitations, not medical guidance or claims of clinical validation.
5. **From transcript to structured output: where an AI pipeline can fail** — map data flow, error handling, provenance, and evaluation boundaries.
6. **Testing asynchronous Python media pipelines: retries, provider failures, and deterministic outputs** — useful if backed by the actual implementation and tests.
7. **Debugging a real integration failure: symptom, root cause, test, and fix** — write a specific post from a genuine, publishable incident or project issue.
8. **A project case study for a React/TypeScript application: architecture, component boundaries, and trade-offs** — only if the project page can add technical specifics beyond the current overview.
9. **How I evaluate AI-generated output before it reaches a user** — explain application QA and review; distinguish this from formal safety/compliance certification.
10. **Responsible vulnerability disclosure or assessment methodology** — publish only if the author has a real authorized process or assessment to document; otherwise omit it for now.

LinkedIn, Medium, and GitHub

LinkedIn: Align headline and About copy with the portfolio’ s actual role and target level. Feature the two strongest project case studies and résumé. Each post should point to a complete project/article page and describe one original engineering decision, test, or limitation; do not copy/paste generic AI trend posts. Use hiring terms naturally rather than stacking every keyword.

Medium: Treat Medium as a distribution channel for a small number of substantive technical articles. Keep the portfolio as the definitive home for project evidence, include a clear author/profile link, and add a canonical link to the portfolio version when cross-posting duplicate article text. Favor article titles that describe the specific engineering question, not broad “AI engineer” keyword targets.

GitHub: Link public repositories only where the code is intended to be shared. Make README files consistent with the portfolio: purpose, role/contribution, setup, architecture, tests, limitations, and demo. The site says some case-study repositories are private; do not suggest otherwise or publish private code. 2 3 4 5

Prioritized 90-day plan

Period	Work	Verification
Days 1–14: technical source fixes	Fix per-route canonical output so original HTML, rendered DOM, sitemap, and internal links agree. Ensure unique route-specific titles/descriptions/H1s in initial HTML where possible. Decide whether `/blog` should remain an alias or become a real index. Inspect mobile blocking resources and JS bundle before optimization.	Compare raw `curl` /view-source and rendered DOM for `/`, `/about`, `/work`, `/work/iscribe`, `/work/rshorts`, and the article route. Check canonical points to the correct `www` URL on each. Confirm sitemap URLs return 200. Run local Lighthouse again on homepage and representative project/article pages.
Days 15–30: establish baseline	Set up or access Google Search Console and verify the domain property; submit/check the sitemap and inspect the key URLs. Capture a 28-day and, if available, 12-month baseline. Do not infer no indexation solely from a public search sample.	Record clicks, impressions, CTR, average position, query/page/country/device dimensions, indexed-page status, Google-selected canonical, and any crawl/indexing errors. Treat small-volume data cautiously.
Days 31–60: strengthen evidence pages	Improve the Work hub’s recruiter path and publish one genuinely new case study or technical article based on real implementation/test artifacts. Add relevant internal links from About, Work, projects, article, résumé, and GitHub. Align LinkedIn Featured items and one Medium distribution post.	Check that target routes render without interaction, titles/canonicals remain correct after JS hydration, links work, and the updated pages appear in URL Inspection. Track visits/referrals separately from organic query demand.
Days 61–90: iterate from evidence	Use Search Console queries and page data to refine page titles/intros and choose one follow-up article. Continue only with content that supports the real career objective. Improve performance based on repeatable lab findings and, if available, field data.	Compare baseline and follow-up clicks/impressions by page and query; separate name/brand searches from non-brand searches. Track resume clicks, GitHub outbound clicks, and contact submissions with consent-aware analytics. Do not set ranking guarantees or treat a

		change in one test run as causal proof.

Measurement plan and baseline requirements

1. **Search Console:** domain verification; submitted sitemap; indexed URL count and URL Inspection for homepage, About, Work, top projects, and article; Google-selected vs user-declared canonical; query/page/country/device clicks, impressions, CTR, and position.
2. **Analytics:** organic landing pages, engaged sessions, contact submissions, résumé clicks, GitHub/LinkedIn outbound clicks, and referral sources. Separate organic search from direct, social, and referral traffic. Keep privacy/consent settings appropriate.
3. **Keyword validation:** after enough impressions accrue, export actual Search Console queries for India and relevant pages. Treat those as first-party impressions/clicks for this property, not total market search volume. If exact market demand is still required, obtain a legitimate keyword-planner or Semrush dataset later and retain the source, country, date range, and metric definition.
4. **Performance:** repeat Lighthouse at consistent settings, then compare with field Core Web Vitals/Search Console or Chrome UX Report if the domain has sufficient eligible data. The current local run is a lab baseline only.
5. **Recruiter value:** annotate links from LinkedIn/Medium to portfolio pages and track qualified contacts/interview conversations; these outcomes need not move with Google rankings and should be reported separately.

Five highest-value opportunities and evidence needed

1. **Resolve route-level canonical mismatch in the original HTML.** Evidence already observed: raw HTML for multiple routes points to the homepage, while browser rendering sets route-specific canonicals. Validate after any future change with source HTML, rendered DOM, sitemap consistency, and Search Console’ s Google-selected canonical for each important route.
2. **Strengthen distinct page signals and indexability for the highest-value pages.** Evidence already observed: route content and rendered titles are distinct, but initial HTML repeats homepage metadata. Validate with raw HTML for unique title/description/H1/canonical on About, Work, iScribe, PromptToShort, and the article; then confirm indexing and impressions per URL.

3. **Measure real, first-party search demand before expanding keyword targeting.** Current evidence gap: no GSC data or credible public volume/KD was available. Validate using Search Console query impressions/clicks by page, India country, and date range; supplement with a dated keyword-tool export only if broader market volume is needed.
4. **Turn the strongest project proof into a clear recruiter evaluation path.** Evidence already observed: the Work hub and project pages contain architecture/testing/scope detail. Validate usefulness through project-page visits from LinkedIn/GitHub, résumé/GitHub clicks, qualified recruiter feedback, and contacts—not assumed ranking benefit.
5. **Improve mobile loading only where repeatable diagnostics justify the work.** Current evidence: one local simulated mobile run reported 4.0 s LCP, 410 ms TBT, and render-blocking/unused-JS opportunities; PageSpeed Insights API was unavailable due quota. Validate with repeat Lighthouse runs and real-user field data before attributing business impact.

References

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